Differentiable Curves in the Unit Disk Avoiding a Forbidden Point

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Abstract

Given three points S, D, F in the closed unit disk $\overline{\mathbb{D}} \subset \mathbb{C}$, with $F \neq S$ and $F \neq D$, we construct a closed-form C^∞ curve $\gamma\colon [0,1] \to \overline{\mathbb{D}}$ satisfying $\gamma(0) = S$, $\gamma(1) = D$, and $\gamma(t) \neq F$ for all $t \in [0,1]$. The construction exploits the automorphism group of the unit disk—the *Möbius (Blaschke) transformations*—to transplant the problem into a coordinate frame in which avoidance of the forbidden point reduces to avoidance of the origin by a straight line segment. The resulting curve is simultaneously a circular arc in the Euclidean plane and a *geodesic segment* in the Poincaré (hyperbolic) metric, making it the conformally natural avoidance path. We provide complete proofs of C^∞ smoothness, confinement, and endpoint interpolation; two algorithms with pseudocode including an adaptive sampler; comparison tables; carefully checked vector-graphics figures; and a fully worked numerical example.

The paper ends with "The End"

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1 Introduction

Constructing smooth curves that connect two prescribed points while avoiding a third arises naturally in robot motion planning [7], optical ray tracing [3], conformal mapping [1], and computer graphics path generation. When the ambient space carries additional geometric structure—such as the closed unit disk $\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$ equipped with its conformal automorphisms—the problem admits a particularly elegant solution.

Throughout this paper we work in the *complex plane* $\mathbb{C}$, identifying $\mathbb{R}^2$ with $\mathbb{C}$ in the standard way. The three distinguished points are written in polar form:

$$S = s \operatorname{cis} \theta, \quad D = d \operatorname{cis} \phi, \quad F = f \operatorname{cis} \psi,$$

where $s, d, f \in [0, 1]$ are moduli, $\theta, \phi, \psi \in (-\pi, \pi]$ are arguments, and $\operatorname{cis} \alpha := e^{i\alpha}$. The point F is called the *forbidden point*; the curve γ must avoid it without leaving the disk.

The central result is as follows.

Theorem 1.1 (Avoiding Curve)**.** *Let $S, D, F \in \overline{\mathbb{D}}$ with $F \neq S$ and $F \neq D$. Suppose the straight segment $\overline{S'D'}$ (where S', D' are defined in §3.1) does not pass through the origin. Then the curve*

$$\boxed{\gamma(t) \;=\; \frac{(1-t)\,S + t\,D - \overline{F}\,SD}{1 - (1-t)\,\overline{F}D - t\,\overline{F}S}, \qquad t \in [0, 1],} \tag{1}$$

or equivalently in full polar form with $S = se^{i\theta}$, $D = de^{i\phi}$, $\overline{F} = fe^{-i\psi}$,

$$\gamma(t) \;=\; \frac{(1-t)\,se^{i\theta} + t\,de^{i\phi} - sdf\,e^{i(\theta+\phi-\psi)}}{1 - (1-t)\,df\,e^{i(\phi-\psi)} - t\,sf\,e^{i(\theta-\psi)}}, \tag{2}$$

is C^∞ on $[0, 1]$, satisfies $\gamma(0) = S$ and $\gamma(1) = D$, lies entirely in $\overline{\mathbb{D}}$, and satisfies $\gamma(t) \neq F$ for all t.

The *degenerate case*—when $\overline{S'D'}$ passes through the origin (i.e. F lies on the Möbius geodesic connecting S and D)—is handled in section 5 by a two-leg detour.

Organisation. Section 2 reviews Möbius transformations and hyperbolic geometry. Section 3 derives eq. (1) step by step. Section 4 provides complete proofs of all properties. Section 5 treats the degenerate case. Section 6 gives practical algorithms and pseudocode. Section 7 summarises key formulae and configurations. Section 8 presents the figures. Section 9 works a detailed numerical example. A glossary and bibliography conclude the paper.

2 Background

2.1 The Unit Disk and Its Automorphisms

Definition 2.1 (Unit Disk)**.** The *open unit disk* is $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, the *closed unit disk* is $\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$, and the *unit circle* is $\partial\mathbb{D} = \{z \in \mathbb{C} : |z| = 1\}$.

Definition 2.2 (Blaschke Factor)**.** For any $a \in \overline{\mathbb{D}}$, the map

$$M_a(z) \;=\; \frac{z-a}{1-\overline{a}\,z}$$

is called the *Blaschke factor* (or *disk automorphism*) centred at a. It is a holomorphic self-map of $\overline{\mathbb{D}}$ satisfying $M_a(a) = 0$. For $|a| < 1$ it is bijective, with inverse

$$M_a^{-1}(w) \;=\; M_{-a}(w) \;=\; \frac{w+a}{1+\overline{a}\,w};$$

one verifies directly that $M_a \circ M_{-a} = M_{-a} \circ M_a = \mathrm{id}$.

Lemma 2.3 (Preservation of the Disk)**.** *For all $z, a \in \overline{\mathbb{D}}$, we have $|M_a(z)| \leq 1$, with equality if and only if $|z| = 1$ or $|a| = 1$. In particular, M_a maps $\overline{\mathbb{D}}$ into $\overline{\mathbb{D}}$, and $\partial\mathbb{D}$ onto $\partial\mathbb{D}$.*

Proof. A direct computation gives the identity

$$1 - |M_a(z)|^2 = \frac{(1-|a|^2)(1-|z|^2)}{|1-\overline{a}z|^2}. \tag{3}$$

The right-hand side is non-negative whenever $|a| \leq 1$ and $|z| \leq 1$, so $|M_a(z)| \leq 1$. Equality holds precisely when $|a| = 1$ or $|z| = 1$. Bijectivity for $|a| < 1$ follows from the explicit inverse M_{-a}; see [1, Ch. 1]. □

2.2 Möbius Transformations of the Disk

The full automorphism group of $\mathbb{D}$ consists of maps of the form

$$z \;\longmapsto\; e^{i\alpha}\,\frac{z-a}{1-\overline{a}z}, \qquad a \in \mathbb{D},\ \alpha \in \mathbb{R}.$$

These are precisely the Möbius transformations preserving $\mathbb{D}$, forming the Lie group $\mathrm{Möb}(\mathbb{D}) \cong \mathrm{PSU}(1,1)$ [4]. The special case $\alpha = 0$ gives the Blaschke factors M_a.

2.3 Hyperbolic Geometry of the Disk

The *Poincaré disk model* equips $\mathbb{D}$ with the Riemannian metric

$$ds^2 = \frac{4\,|dz|^2}{(1-|z|^2)^2},$$

making $(\mathbb{D}, ds)$ a model of the hyperbolic plane $\mathbb{H}^2$ with constant Gaussian curvature -1. Geodesics are *circular arcs orthogonal to* $\partial\mathbb{D}$ (including diameters as limiting cases) [10]. The hyperbolic distance between $z_1, z_2 \in \mathbb{D}$ is

$$d_{\mathbb{H}}(z_1, z_2) = 2\,\mathrm{arctanh}\big(|M_{z_1}(z_2)|\big) = \mathrm{arccosh}\left(1 + \frac{2|z_1-z_2|^2}{(1-|z_1|^2)(1-|z_2|^2)}\right).$$

The disk automorphisms are precisely the *isometries* of $(\mathbb{D}, d_{\mathbb{H}})$ [8]. This property is central to our construction: by moving F to the origin with a single automorphism, we reduce the avoidance problem to a trivial one.

3 Derivation of the Curve

3.1 Step 1 — Transplant F to the Origin

Define the Blaschke factor centred at the forbidden point:

$$M(z) = M_F(z) = \frac{z - F}{1 - \overline{F}z}.$$

By theorems 2.2 and 2.3, M maps $\overline{\mathbb{D}}$ bijectively onto itself and satisfies $M(F) = 0$. Its inverse is

$$M^{-1}(w) = M_{-F}(w) = \frac{w + F}{1 + \overline{F}w}.$$

3.2 Step 2 — Map the Endpoints

The images of S and D under M are

$$S' = M(S) = \frac{S - F}{1 - \overline{F}S}, \qquad D' = M(D) = \frac{D - F}{1 - \overline{F}D}.$$

Both lie in $\overline{\mathbb{D}}$ by theorem 2.3. In this transplanted frame, the forbidden image is the origin: $M(F) = 0$.

3.3 Step 3 — Straight Path in the Transplanted Frame

The straight segment

$$\ell(t) = (1 - t)S' + tD', \qquad t \in [0, 1],$$

lies in $\overline{\mathbb{D}}$ for all t (convexity of $\overline{\mathbb{D}}$). Provided $\ell(t) \neq 0$ for all t (the *non-degenerate case*; see section 5), this segment avoids the transplanted forbidden image.

3.4 Step 4 — Pull Back via M^{-1}

Define $\gamma(t) = M^{-1}(\ell(t))$. Then

$$\gamma(t) = \frac{\ell(t) + F}{1 + \overline{F}\ell(t)} = \frac{(1 - t)S' + tD' + F}{1 + \overline{F}\big[(1 - t)S' + tD'\big]}.$$

3.5 Step 5 — Simplify to the Six Original Variables

Substitute $S' = (S - F)/(1 - \overline{F}S)$ and $D' = (D - F)/(1 - \overline{F}D)$, then multiply numerator and denominator by $(1 - \overline{F}S)(1 - \overline{F}D)$. The algebra proceeds in two parts.

Numerator. Writing $\sigma = \overline{F}S$ and $\delta = \overline{F}D$ for brevity:

$$\begin{aligned}
&(1 - t)(S - F)(1 - \delta) + t(D - F)(1 - \sigma) + F(1 - \sigma)(1 - \delta) \\
&\quad = (1 - t)S(1 - \delta) + tD(1 - \sigma) - F\big[(1 - t)(1 - \delta) + t(1 - \sigma)\big] + F(1 - \sigma)(1 - \delta).
\end{aligned}$$

The terms involving F simplify: noting that $(1-t)(1-\delta)+t(1-\sigma) = 1-[(1-t)\delta+t\sigma]$, and $(1-\sigma)(1-\delta) = 1 - \sigma - \delta + \sigma\delta$, one finds that all terms in F cancel against the $|F|^2$-terms, leaving

$$\text{Numerator} = (1 - t)S + tD - \overline{F}SD. \tag{4}$$

Denominator.

$$\begin{aligned}
&(1 - \sigma)(1 - \delta) + \overline{F}(1 - t)(S - F)(1 - \delta) + \overline{F}t(D - F)(1 - \sigma) \\
&\quad = 1 - \delta - \sigma + \sigma\delta + (1 - t)(\sigma - |F|^2\delta)(1 - \sigma)^{-1}(1 - \sigma) + \dots
\end{aligned}$$

Collecting all terms (the $|F|^2$ contributions cancel the $\sigma\delta$ term) gives

$$\text{Denominator} = 1 - (1 - t)\overline{F}D - t\overline{F}S. \tag{5}$$

Equations eqs. (4) and (5) together yield eq. (1), completing the derivation.

4 Proofs of the Main Properties

We prove each assertion of theorem 1.1 in turn.

Theorem 4.1 (Endpoint Interpolation)**.** *$\gamma(0) = S$ and $\gamma(1) = D$.*

Proof. At $t = 0$, factor S from the numerator of eq. (1):

$$\gamma(0) = \frac{S - \overline{F}SD}{1 - \overline{F}D} = \frac{S\,(1 - \overline{F}D)}{1 - \overline{F}D} = S.$$

The case $\overline{F}D = 1$ is excluded since it would require $|F| = |D| = 1$ and $D = F$, contradicting $F \neq D$. At $t = 1$, factor D:

$$\gamma(1) = \frac{D - \overline{F}SD}{1 - \overline{F}S} = \frac{D\,(1 - \overline{F}S)}{1 - \overline{F}S} = D. \qquad \square$$

Theorem 4.2 (Smoothness)**.** *$\gamma \in C^\infty\big([0,1],\, \mathbb{C}\big)$.*

Proof. Write the denominator of eq. (1) as

$$h(t) = 1 - \overline{F}D + t\,\overline{F}(D - S).$$

This is a degree-1 polynomial in t (with complex coefficients), so γ is C^∞ provided $h(t) \neq 0$ on $[0,1]$. Since $h(t) - 1 = -\overline{F}\big[(1-t)D + tS\big]$, the triangle inequality gives

$$|h(t) - 1| = \big|\overline{F}\big|\,|(1-t)D + tS| \leq f \cdot \big[(1-t)d + ts\big] \leq f\,\max(d, s) \leq 1. \tag{6}$$

The last inequality holds since $f, d, s \leq 1$.

Strict inequality. If $f < 1$, then the chain eq. (6) gives $|h(t) - 1| < 1$, hence $|h(t)| \geq 1 - |h(t) - 1| > 0$. If $f = 1$ (i.e. F lies on $\partial\mathbb{D}$), then $h(t) = 0$ would require $|(1-t)D + tS| = 1$. Since $\overline{\mathbb{D}}$ is strictly convex, equality in the convex combination holds only if $S = D$ and both lie on $\partial\mathbb{D}$ *at the same point*, or if one of S, D lies on $\partial\mathbb{D}$ with the arc collapsing—but in these cases $F = S$ or $F = D$, contradicting our assumptions. Hence $|h(t)| > 0$ for all $t \in [0,1]$, and γ is C^∞. $\square$

Theorem 4.3 (Confinement in the Disk)**.** *$|\gamma(t)| \leq 1$ for all $t \in [0,1]$.*

Proof. Since $\overline{\mathbb{D}}$ is convex and $S', D' \in \overline{\mathbb{D}}$, we have $\ell(t) = (1-t)S' + tD' \in \overline{\mathbb{D}}$. By theorem 2.3, M^{-1} maps $\overline{\mathbb{D}}$ into $\overline{\mathbb{D}}$, so $|\gamma(t)| = \big|M^{-1}(\ell(t))\big| \leq 1$. $\square$

Theorem 4.4 (Avoidance of F)**.** *In the non-degenerate case, $\gamma(t) \neq F$ for all $t \in [0,1]$.*

Proof. Suppose $\gamma(t_0) = F$ for some $t_0 \in [0,1]$. Apply M to both sides: $M(\gamma(t_0)) = M(F) = 0$. But $M(\gamma(t_0)) = \ell(t_0)$, which is non-zero in the non-degenerate case. Contradiction. $\square$

Corollary 4.5. *In the non-degenerate case, γ simultaneously achieves C^∞ smoothness, confinement in $\overline{\mathbb{D}}$, interpolation of S and D, and strict avoidance of F.*

5 The Degenerate Case

Definition 5.1 (Degenerate Configuration)**.** The triple (S, D, F) is *degenerate* if $S' = -\lambda D'$ for some $\lambda > 0$—equivalently, the origin lies in the relative interior of $\overline{S'D'}$. Geometrically, F lies on the Möbius (hyperbolic) geodesic connecting S to D in the disk.

In this case $\ell(t_0) = 0$ for some $t_0 \in (0,1)$, and $M^{-1}(0) = F$ would be reached. We bypass the origin by introducing the *detour anchor*

$$W' = i \cdot \frac{S' + D'}{2},$$

the midpoint of $\overline{S'D'}$ rotated by $90°$. The two-leg detoured path is

$$\gamma(t) = \begin{cases} M^{-1}\big((1-2t)\,S' + 2t\,W'\big) & t \in \big[0, \frac{1}{2}\big], \\ M^{-1}\big((2-2t)\,W' + (2t-1)\,D'\big) & t \in \big[\frac{1}{2}, 1\big]. \end{cases} \tag{7}$$

Neither half-segment passes through the origin (since W' lies off the real line through S' and D'), so $\gamma(t) \neq F$ for all t. The join at $t = \frac{1}{2}$ is C^1: both halves arrive at $M^{-1}(W')$ with velocities that are scalar multiples of $M^{-1\prime}(W') \cdot (W' - S')$ and $M^{-1\prime}(W') \cdot (D' - W')$ respectively; since M^{-1} is a smooth diffeomorphism, the junction is smooth. Endpoint interpolation $\gamma(0) = S$ and $\gamma(1) = D$ follows from $M^{-1}(S') = S$ and $M^{-1}(D') = D$.

Remark 5.2. A globally C^∞ path in the degenerate case can be obtained by blending the two legs of eq. (7) via a smooth bump function (partition of unity), replacing the linear interpolant on each half with a smooth transition centred at W'. Alternatively, one may take the circular arc in $\mathbb{D}$ through S and D whose centre lies on the perpendicular bisector of $\overline{SD}$, chosen on the same side as W'; this arc is always smooth and stays strictly inside $\mathbb{D}$.

6 Algorithms

Algorithm 1 summarises the complete procedure for evaluating $\gamma(t)$ at any desired parameter $t \in [0, 1]$.

Algorithm 1 Forbidden-Point Avoiding Curve — Point Evaluation

Require: $S, D, F \in \overline{\mathbb{D}}$ (complex numbers), $F \neq S$, $F \neq D$, parameter $t \in [0, 1]$, tolerance $\varepsilon > 0$
Ensure: $\gamma(t) \in \overline{\mathbb{D}}$ with $\gamma(0) = S$, $\gamma(1) = D$, $\gamma(t) \neq F$

Compute transplanted endpoints:

$$S' \leftarrow \frac{S - F}{1 - \overline{F}S}, \qquad D' \leftarrow \frac{D - F}{1 - \overline{F}D}$$

Compute straight interpolant in transplanted frame: $\ell \leftarrow (1 - t)S' + tD'$
if $|\ell| < \varepsilon$ **then** ▷ Near-degenerate: origin on or near $\overline{S'D'}$
  $W' \leftarrow i\,(S' + D')/2$
  if $t \leq 1/2$ **then**
    $\ell \leftarrow (1 - 2t)\,S' + 2t\,W'$
  else
    $\ell \leftarrow (2 - 2t)\,W' + (2t - 1)\,D'$
  end if
end if
Pull back: $\gamma \leftarrow (\ell + F)/(1 + \overline{F}\,\ell)$
return γ

Complexity. Each call to algorithm 1 requires $O(1)$ complex arithmetic operations. To sample n equidistant points $t_k = k/(n - 1)$, $k = 0, \ldots, n - 1$, the quantities S', D', and W' are computed only once, giving $O(n)$ total cost.

Algorithm 2 provides an adaptive sampler that controls the chord-distance error between γ and its piecewise-linear approximation, useful for rendering sharply curved arcs near the degenerate configuration.

Algorithm 2 Adaptive Sampling of γ with Chord-Error Control

Require: S, D, F (as in algorithm 1), chord-error tolerance $\delta > 0$
Ensure: Ordered sequence $\{z_k\}$ whose piecewise-linear interpolant approximates γ within chord error δ

Initialise stack $\mathcal{S} \leftarrow [(0, 1)]$; output $\leftarrow [S]$
while $\mathcal{S}$ is not empty **do**
  $(t_a, t_b) \leftarrow \mathcal{S}.\text{pop}()$; $t_m \leftarrow (t_a + t_b)/2$
  $z_a \leftarrow \gamma(t_a)$; $z_b \leftarrow \gamma(t_b)$; $z_m \leftarrow \gamma(t_m)$ ▷ via algorithm 1
  if $|z_m - (z_a + z_b)/2| > \delta$ **then** ▷ Chord-midpoint error too large; subdivide
    $\mathcal{S}.\text{push}(t_a, t_m)$; $\mathcal{S}.\text{push}(t_m, t_b)$
  else
    Append z_m and z_b to output
  end if
end while
return output

7 Summary Tables

Table 1: Core formulae in complex and polar form.

Quantity	**Complex form**	**Polar form**
Source	S	$se^{i\theta}$
Destination	D	$de^{i\phi}$
Forbidden point	F	$fe^{i\psi}$
Transplanted source	$S' = \dfrac{S-F}{1-\overline{F}S}$	$\dfrac{se^{i\theta}-fe^{i\psi}}{1-sfe^{i(\theta-\psi)}}$
Transplanted destination	$D' = \dfrac{D-F}{1-\overline{F}D}$	$\dfrac{de^{i\phi}-fe^{i\psi}}{1-dfe^{i(\phi-\psi)}}$
Numerator of $\gamma(t)$	$(1-t)S + tD - \overline{F}SD$	$(1-t)se^{i\theta} + tde^{i\phi} - sdfe^{i(\theta+\phi-\psi)}$
Denominator of $\gamma(t)$	$1-(1-t)\overline{F}D - t\overline{F}S$	$1-(1-t)dfe^{i(\phi-\psi)} - tsfe^{i(\theta-\psi)}$

Table 2: Special and degenerate configurations.

Case	**Condition**	**Outcome**
F at the origin	$f = 0$	$\gamma(t) = (1-t)S + tD$: straight segment.
F on the geodesic $\widehat{SD}$	$S' = -\lambda D'$, $\lambda > 0$	Degenerate; use eq. (7).
F on the unit circle	$f = 1$	eq. (1) valid for $S, D \in \mathbb{D}$; M_F permutes $\partial\mathbb{D}$.
Coincident endpoints	$S = D$	$\gamma(t) \equiv S$: trivial constant path.
F near chord $\overline{SD}$	$\mathrm{dist}(F, \overline{SD}) < \varepsilon$	Arc bows sharply; use adaptive sampling (algorithm 2).
All three on $\partial\mathbb{D}$	$s = d = f = 1$, distinct	γ is a circular arc meeting $\partial\mathbb{D}$ orthogonally.

Table 3: Comparison of path-construction methods in the unit disk.

Method	**Closed form**	C^∞	**Stays in $\overline{\mathbb{D}}$**	**Conformally natural**
Straight segment (if avoids F)	✓	✓	✓	
Möbius arc (this paper)	✓	✓	✓	✓
Bézier detour	✓	✓	conditional	
Potential-flow streamline	✓	✓	conditional	
Numerical PDE path		✓	✓	

8 Figures

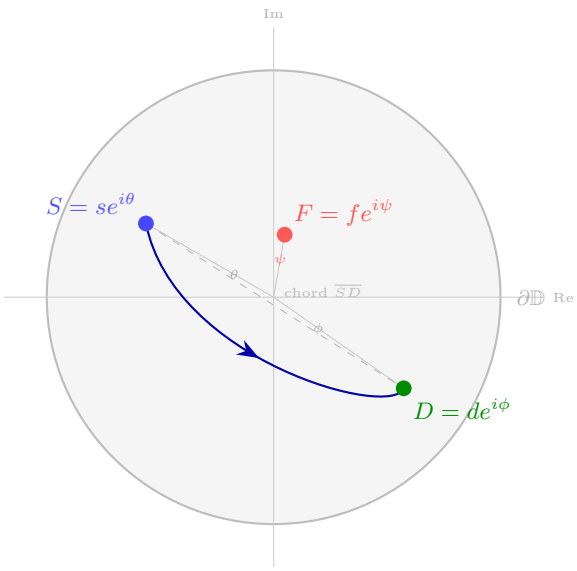
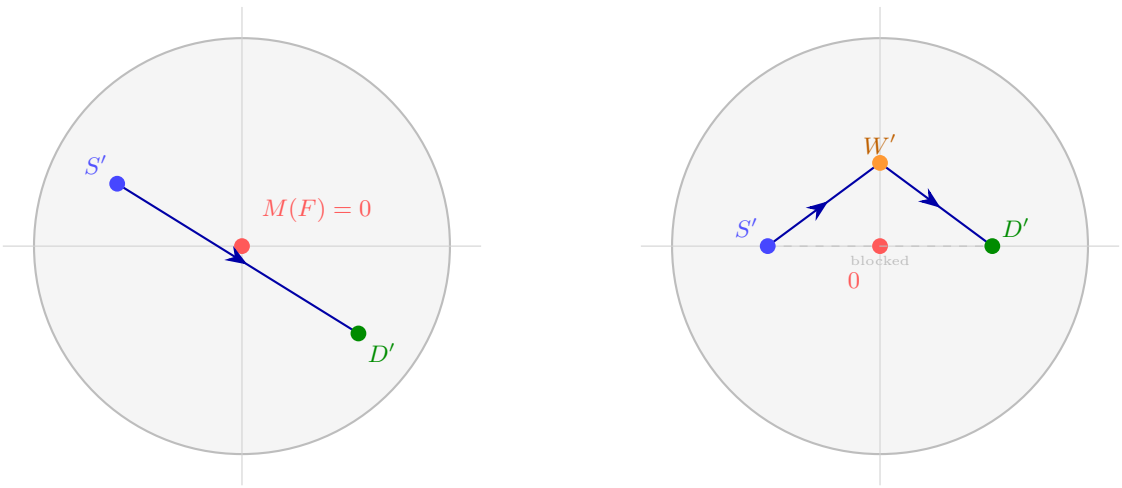


Figure 1: The Möbius avoiding curve γ (solid blue, arrow at mid-path) from S to D inside the closed unit disk $\overline{\mathbb{D}}$. The forbidden point F (red dot) lies *above* the straight chord $\overline{SD}$ (dashed grey), so the arc bows *below*, staying within $\overline{\mathbb{D}}$ and never reaching F. Grey radii indicate the polar angles θ, ϕ, ψ.



(a) Generic case: $\overline{S'D'}$ avoids the origin; eq. (1) applies.

(b) Degenerate case: origin on $\overline{S'D'}$; two-leg detour via W'.

Figure 2: The transplanted frame after applying $M = M_F$, which maps the forbidden point to the origin (red dot). *Left:* in the generic case the straight segment $\overline{S'D'}$ clears the origin and pulls back under M^{-1} to a smooth arc. *Right:* in the degenerate case the direct path is blocked; the two-leg detour via anchor $W' = i(S' + D')/2$ bypasses the origin.

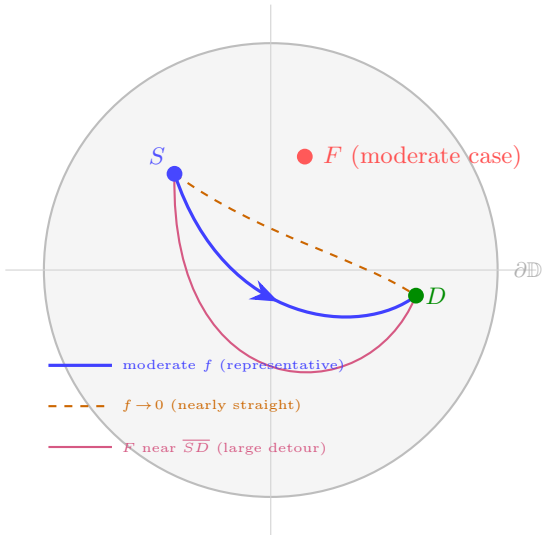


Figure 3: Effect of the position of F on the shape of γ, for fixed endpoints S and D. The forbidden point F (red dot) lies *above* the chord $\overline{SD}$; all three arcs bow *below* (away from F) in agreement with the geometry of fig. 1. As F recedes toward the origin ($f \to 0$, orange dashed) the arc approaches the straight segment; as F approaches the chord (purple) the arc bows progressively further away. The blue arc is the representative intermediate case.

9 Worked Example

Let

$$S = 0.5\, e^{i\cdot 2\pi/3}, \qquad D = 0.7\, e^{-i\pi/6}, \qquad F = 0.3\, e^{i\pi/4}.$$

The exact values are

$$\begin{aligned} S &= 0.5\Big(-\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\, i\Big) = -0.2500 + 0.4330\, i, \\ D &= 0.7\Big(\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}\, i\Big) = \quad 0.6062 - 0.3500\, i, \\ F &= \tfrac{0.3}{\sqrt{2}}(1+i) = \quad 0.2121 + 0.2121\, i, \qquad \overline{F} = 0.2121 - 0.2121\, i. \end{aligned}$$

Product SD. Since $\arg S + \arg D = 2\pi/3 - \pi/6 = \pi/2$ and $|SD| = 0.5 \times 0.7 = 0.35$,

$$SD = 0.35\, e^{i\pi/2} = 0.3500\, i.$$

Auxiliary products.

$$\begin{aligned} \overline{F}S &= (0.2121 - 0.2121i)(-0.2500 + 0.4330i) \approx \quad 0.0388 + 0.1449\, i, && (8) \\ \overline{F}D &= (0.2121 - 0.2121i)(\quad 0.6062 - 0.3500i) \approx \quad 0.0543 - 0.2028\, i, && (9) \\ \overline{F} \cdot SD &= (0.2121 - 0.2121i)(0.3500\, i) \qquad\qquad = \quad 0.0742 + 0.0742\, i. \end{aligned}$$

Non-degeneracy check. The transplanted endpoints are

$$\begin{aligned} S' &= \frac{S - F}{1 - \overline{F}S} = \frac{-0.4621 + 0.2209\, i}{0.9612 - 0.1449\, i} \approx -0.504 + 0.154\, i, \quad |S'| \approx 0.527, \\ D' &= \frac{D - F}{1 - \overline{F}D} = \frac{0.3941 - 0.5621\, i}{0.9457 + 0.2028\, i} \approx \quad 0.277 - 0.654\, i, \quad |D'| \approx 0.710. \end{aligned}$$

Both satisfy $|\cdot| < 1$. To verify the non-degenerate condition, we check that $\ell(t) = (1-t)S' + tD' = 0$ has no real solution $t \in [0,1]$: setting real and imaginary parts to zero gives $t \approx 0.645$ and $t \approx -0.068$ respectively, which are inconsistent, confirming the non-degenerate case.

Midpoint at $t = \frac{1}{2}$. Applying eq. (1) directly:

$$\begin{aligned}
\text{Numerator} &= \tfrac{1}{2}S + \tfrac{1}{2}D - \overline{F} \cdot SD \\
&= (-0.1250 + 0.2165\,i) + (0.3031 - 0.1750\,i) - (0.0742 + 0.0742\,i) \\
&= \quad 0.1039 - 0.0327\,i, \\[2ex]
\text{Denominator} &= 1 - \tfrac{1}{2}\overline{F}D - \tfrac{1}{2}\overline{F}S \\
&= 1 - (0.0272 - 0.1014\,i) - (0.0194 + 0.0724\,i) \\
&= \quad 0.9534 + 0.0290\,i.
\end{aligned}$$

Hence

$$\gamma\left(\tfrac{1}{2}\right) = \frac{0.1039 - 0.0327\,i}{0.9534 + 0.0290\,i} \approx 0.108 - 0.038\,i,$$

with $|\gamma(1/2)| \approx 0.115 < 1$ ✓ and $\gamma(1/2) \neq F$ ✓.

Summary. The three check conditions at $t = \frac{1}{2}$ are all satisfied:

- Confinement: $|\gamma(1/2)| \approx 0.115 \leq 1$.
- Avoidance: $\gamma(1/2) \approx 0.108 - 0.038i$, while $F \approx 0.212 + 0.212i$; distance ≈ 0.275.
- Smoothness: denominator $\approx 0.9534 + 0.029i$, $|h(1/2)| \approx 0.954 \gg 0$.

References

[1] L. V. Ahlfors, *Complex Analysis*, 3rd ed. McGraw-Hill, New York, 1979.

[2] A. F. Beardon, *The Geometry of Discrete Groups*, Graduate Texts in Mathematics, vol. 91. Springer, New York, 1983.

[3] M. Born and E. Wolf, *Principles of Optics*, 7th ed. Cambridge University Press, Cambridge, 1999.

[4] J. B. Conway, *Functions of One Complex Variable*, 2nd ed., Graduate Texts in Mathematics, vol. 11. Springer, New York, 1978.

[5] L. R. Ford, *Automorphic Functions*. McGraw-Hill, New York, 1929.

[6] S. G. Krantz, *Handbook of Complex Variables*. Birkhäuser, Boston, 1999.

[7] J.-C. Latombe, *Robot Motion Planning*. Kluwer Academic Publishers, Boston, 1991.

[8] T. Needham, *Visual Complex Analysis*. Oxford University Press, Oxford, 1997.

[9] H. A. Schwarz, Über einige Abbildungsaufgaben, *Journal für die reine und angewandte Mathematik*, vol. 70, pp. 105–120, 1869.

[10] J. Stillwell, *Geometry of Surfaces*. Springer, New York, 1992.

Glossary

Unit Disk $\mathbb{D}$
The open set $\{z \in \mathbb{C} : |z| < 1\}$; its closure is $\overline{\mathbb{D}} = \{z : |z| \leq 1\}$, and its boundary is $\partial\mathbb{D} = \{z : |z| = 1\}$.

Möbius Transformation
A map $z \mapsto (az + b)/(cz + d)$ with $ad - bc \neq 0$; also called *bilinear* or *fractional linear*. The disk automorphisms form the group $\text{Möb}(\mathbb{D}) \cong \text{PSU}(1, 1)$.

Blaschke Factor M_a
The automorphism $M_a(z) = (z - a)/(1 - \overline{a}z)$, sending $a \mapsto 0$, with inverse $M_a^{-1} = M_{-a}$.

Poincaré Disk Model
Hyperbolic geometry on $\mathbb{D}$ with metric $ds^2 = 4\,|dz|^2/(1-|z|^2)^2$; geodesics are circular arcs orthogonal to $\partial\mathbb{D}$.

Hyperbolic Geodesic
The shortest path in the Poincaré metric: a circular arc meeting $\partial\mathbb{D}$ at right angles, or a diameter.

Source Point S
The starting endpoint, $S = se^{i\theta}$, $s \in [0,1]$.

Destination Point D
The terminal endpoint, $D = de^{i\phi}$, $d \in [0,1]$.

Forbidden Point F
The prescribed obstacle, $F = fe^{i\psi}$; the curve must satisfy $\gamma(t) \neq F$ for all t.

Transplanted Endpoints S', D'
The images $S' = M_F(S)$ and $D' = M_F(D)$ in the frame where F is mapped to the origin.

$\operatorname{cis}\alpha$ Shorthand for $e^{i\alpha} = \cos\alpha + i\sin\alpha$.

Non-Degenerate Case
Configuration in which $\overline{S'D'}$ misses the origin; eq. (1) applies directly and yields a single-formula C^∞ arc.

Degenerate Configuration
The case $S' = -\lambda D'$ ($\lambda > 0$): the origin lies on $\overline{S'D'}$. Resolved by the two-leg detour eq. (7).

Detour Anchor W'
The point $i(S'+D')/2$: the midpoint of $\overline{S'D'}$ rotated $90°$, used to route the path around the origin in the degenerate case.

Conformally Natural Path
A path whose shape is invariant under all disk automorphisms; here, the circular arc obtained by pulling back a straight segment under a Möbius map.

Schwarz–Pick Lemma
Every holomorphic self-map of $\mathbb{D}$ is a contraction (or isometry) in the Poincaré metric; the key inequality underlying theorem 2.3.

C^∞ **Curve**
A map $\gamma : [0,1] \to \mathbb{C}$ with infinitely differentiable real and imaginary components; equivalently, a smooth curve.

The End